

Lesson 3 Notes · Functions

Module SC101 · Introduction to Python · Social Coding SA

Why functions?

A function is a named recipe: write the steps once, use them whenever you like. It keeps programs short, readable, and easy to fix — change the recipe in one place and every caller gets the fix.

```
def greet(name):  
    return f"Sawubona, {name}!"  
  
print(greet("Zanele"))    # Sawubona, Zanele!  
print(greet("Tebogo"))    # Sawubona, Tebogo!
```

The parts

def starts the definition · the **parameters** in brackets are the ingredients · the indented body is the recipe · **return** hands the answer back. A function that returns nothing gives back `None`.

```
def average(a, b, c):  
    return (a + b + c) / 3  
  
term_avg = average(65, 72, 80)    # 72.33...
```

Mini-exercises

1. Write `square(n)` that returns $n \times n$. 2. Write `is_pass(mark)` that returns `True` for 50+. 3. Write `band(mark)` returning "Distinction" / "Pass" / "Fail". 4. Reuse `band()` in a loop over [42, 55, 81, 68].